Smoothing the Grünbaum–Sreedharan Stanley–Reisner threefold by strict differential graded Lie algebra methods

Abstract

The Stanley–Reisner threefold of the Grünbaum–Sreedharan eight-vertex triangulated three-sphere is the special fibre of a flat projective scheme over $\mathbf{Q}[[s]]$ whose geometric generic fibre is smooth. We prove this using only a Tate resolution $P \to A$, the strict differential graded Lie algebra of weight-zero S-linear derivations of P, the ordinary Maurer–Cartan equation $\partial\alpha + \frac{1}{2}[\alpha, \alpha] = 0$, elementary obstruction theory in the complete filtered dg Lie algebra $s\,\mathrm{Der}_S(P, P)_0[[s]]$ with the strict Bianchi identity, and exact rational computer algebra. No structure transferred to cohomology, and no operation of arity greater than two, occurs anywhere. The deformation-theoretic bridges are constructed by hand rather than quoted: the certified exact family over $\mathbf{Q}[s]/(s^4)$ is lifted to an explicit square-zero perturbation of the Tate differential, and the all-orders family is read off as the sixteen literal cubic differentials of the degree-one Tate generators. The generic fibre is shown to be Cohen–Macaulay and equidimensional by a Betti-number semicontinuity argument, after which the 280 exact incidence certificates force smoothness.

Contents

1 Statement, data, and conventions

1.1 The special fibre

Put $k = \mathbf{Q}$ and $S = k[x_1, \ldots, x_8]$ with its standard grading. Let Δ be the simplicial complex on $\{1, \ldots, 8\}$ with the twenty tetrahedral facets

$$\begin{gathered}1234,\ 1237,\ 1267,\ 1347,\ 1567,\ 2345,\ 2367,\ 3467,\ 3456,\ 4567,\\ 2358,\ 2368,\ 3568,\ 1268,\ 1568,\ 1248,\ 2458,\ 1478,\ 1578,\ 4578,\end{gathered}$$

the Grünbaum–Sreedharan triangulation [3]. Its Stanley–Reisner ideal is generated by the sixteen ordered monomials

$$\begin{aligned}I = (\, m_1, \ldots, m_{16}\,) = (\, & x_6x_7x_8,\ x_4x_6x_8,\ x_3x_7x_8,\ x_3x_5x_7,\ x_3x_4x_8,\ x_2x_7x_8,\\ & x_2x_5x_7,\ x_2x_5x_6,\ x_2x_4x_7,\ x_2x_4x_6,\ x_1x_4x_6,\ x_1x_4x_5,\\ & x_1x_3x_8,\ x_1x_3x_6,\ x_1x_3x_5,\ x_1x_2x_5\,) \subset S.\end{aligned} \tag{1.1}$$

Set $A = S/I$ and $X_0 = \operatorname{Proj} A \subset \mathbf{P}^7_k$. For $n \geq 1$ write

$$R_n = k[s]/(s^n), \qquad R = k[[s]], \qquad K = k((s)).$$

Theorem 1.1 (smoothing theorem)**.** *There exist sixteen homogeneous cubics $F_1, \ldots, F_{16} \in R[x_1, \ldots, x_8]$ with $F_i \equiv m_i \pmod{s}$, whose reductions modulo s^3 are exactly the sixteen rows of the frozen file* `../referee-packet/data/universal_2jet_QQ.txt`*, such that*

$$X = \operatorname{Proj}\big(R[x_1, \ldots, x_8]/(F_1, \ldots, F_{16})\big) \longrightarrow \operatorname{Spec} R$$

is flat and projective, has special fibre X_0, and has smooth geometric generic fibre. In particular X_0 is smoothable in characteristic zero.

This is the entire claim. The proof does not assert smoothness of the total space, irreducibility of the generic fibre, triviality of its canonical bundle, or identification with any named threefold, and no such assertion is used.

Verified computation 1.2 (special fibre)**.** The exact rational replay of `../referee-packet/code/verify_combinatorics.py` confirms: Δ has 8 vertices, 20 facets, 97 faces, is pure of dimension 3, and its inclusion-minimal nonfaces are exactly the sixteen triples in (1.1); and for every face σ (including $\varnothing$) the reduced rational homology of $\operatorname{lk}_\Delta \sigma$ vanishes below the top dimension, by exact `Fraction` row reduction of boundary matrices. By Reisner's criterion [4], A is Cohen–Macaulay; its Krull dimension is $\dim \Delta + 1 = 4$, so $\operatorname{depth}_{S_+} A = 4$.

1.2 Gradings and sign conventions

Every object below carries two gradings.

Homological degree. P (defined in Section 2) is concentrated in homological degrees ≥ 0 and its differential d has degree -1. A *derivation of cohomological degree* p lowers homological degree by p.

Internal weight. Each x_i has weight 1; every Tate generator carries the weight of the cycle it kills (e_i: weight 3; f_j: weight 4; g_a: weights 5 and 6). Differentials and all derivations used below preserve weight; a subscript 0 denotes the weight-preserving part. The parameter s has weight 0.

For homogeneous derivations θ, η of cohomological degrees p, q set

$$\partial\theta = [d, \theta] = d\theta - (-1)^p \theta d, \qquad [\theta, \eta] = \theta\eta - (-1)^{pq} \eta\theta. \tag{1.2}$$

The graded commutator of derivations is a derivation, $\partial^2 = 0$, and ∂ is a degree-one derivation of the bracket, so (1.2) defines strict dg Lie algebras. These two operations, the ordinary Maurer–Cartan equation, and no other algebraic operation, are the deformation machinery of this paper.

1.3 How every assertion is certified

Each substantive statement below is one of: (i) proved here from the definitions; (ii) a *Verified computation*, meaning an exact rational computer-algebra calculation in the frozen packets of this repository, replayed for this document (the replay commands are in `REPRODUCE.md`); (iii) a classical theorem cited with its hypotheses checked in the text. The classical inputs are: Reisner's criterion [4]; Tate's construction of acyclic closures [8, 1] (re-proved inline as Lemma 2.1); the local flatness criterion [6, Tag 00MK] (re-proved inline in the Artinian case used); the Auslander–Buchsbaum formula and unmixedness of Cohen–Macaulay local rings [2]; Hilbert–Serre dimension theory [2]; the valuative criterion of properness; and the finiteness of the integral closure of a complete discrete valuation ring in a finite extension of its fraction field [5, Ch. II]. No result about structures transferred to cohomology, of any arity, is cited or used.

2 The Tate resolution and the two derivation algebras

2.1 The resolution

Lemma 2.1 (staged Tate resolution)**.** *There is a graded-commutative dg S-algebra P, free as a graded commutative algebra on finitely many generators in each homological degree ≥ 1, with weight-preserving differential d of homological degree -1, such that $P_0 = S$, $H_0(P) = A$, and $H_i(P) = 0$ for $i > 0$. It may be chosen so that its generators in homological degrees* 1, 2, 3 *are*

generators	homological degree	weight
$e_1, \ldots, e_{16}$	1	3
$f_1, \ldots, f_{30}$	2	4
$g_1, \ldots, g_{136}$	3	5 (16 of them), 6 (120 of them),

(2.1)

with $d(e_i) = m_i$, with $d(f_j)$ the columns of an explicit matrix R_0 over S, and with $d(g_a)$ the columns of an explicit matrix Z written in the degree-two basis $\{f_j\} \cup \{e_i e_j\}$; the matrices are the frozen certified data of Computation 2.2 below.

Proof. This is Tate's construction [8, 1]; over $\mathbf{Q}$ no divided powers are needed and we spell it out. Start from $P^{(1)} = S[e_1, \ldots, e_{16}]$ with $d(e_i) = m_i$; then $H_0 = A$. Given $P^{(q)}$ with $H_i(P^{(q)}) = 0$ for $1 \leq i \leq q-1$, the cycle module $Z_q(P^{(q)})$ is a finitely generated graded S-module, because $P_q^{(q)}$ is a finitely generated free S-module and S is Noetherian. Choose finitely many weight-homogeneous cycles $z_1, \dots, z_r$ whose classes generate $H_q(P^{(q)})$, adjoin free (graded-commutative) generators $w_1, \dots, w_r$ of homological degree $q+1$ and the corresponding weights, and set $d(w_b) = z_b$. Every monomial involving some w_b has homological degree $\geq q+1$, so $(P^{(q+1)})_i = (P^{(q)})_i$ for $i \leq q$: hence cycles and boundaries in degrees $< q$ are unchanged, so $H_i(P^{(q+1)}) = 0$ for $1 \leq i < q$, while in degree q the cycles are unchanged and the new boundaries $d(w_b) = z_b$ generate $H_q(P^{(q)})$, so $H_q(P^{(q+1)}) = 0$. Set $P = \bigcup_q P^{(q)}$; each element has fixed homological degree, so $H_i(P) = H_i(P^{(i+2)}) = 0$ for $i \geq 1$ and $H_0(P) = A$. The differential preserves weight because each z_b was chosen weight-homogeneous.

For the specific choice (2.1): the sixteen m_i generate I; Computation 2.2 certifies that the frozen 30-column matrix R_0 generates the full first syzygy module of $(m_1, \dots, m_{16})$ and that the frozen 136 columns, together with the boundaries of the decomposable degree-three elements, generate the full degree-two cycle module of the stage $S[e,f]$. Hence the construction may take exactly these generators in degrees 2 and 3 and continue with generators of degree ≥ 4. □

Verified computation 2.2 (certified low Tate stages)**.** Let D_1, D_2 be the degree-one and degree-two differentials of the algebra on (2.1): D_1 is the row $(m_1, \dots, m_{16})$; D_2 has the 30 columns of R_0 and the 120 columns $d(e_i e_j) = m_i e_j - m_j e_i$. Let D_3 be the matrix of the boundaries of the $480 + 560$ decomposables $e_i f_a$, $e_i e_j e_k$, and let Z be the 150×136 matrix of the frozen $d(g_a)$. The independent audit `../audits/tate-stage/` proves, by two complete homogeneous syzygy computations (Macaulay2) and an independent reconstruction with its own full syzygy computations (Singular), the exact graded module equalities

$$D_1 D_2 = 0, \quad D_2 D_3 = 0, \quad D_2 Z = 0, \quad \ker D_1 = \operatorname{im} D_2, \quad \ker D_2 = \operatorname{im}[D_3\ Z]. \tag{2.2}$$

These are unbounded module identities over S, not finite-degree rank tests. The certificate is `../audits/tate-stage/MATHEMATICAL_CERTIFICATE.md`.

Remark 2.3 (exactly what the 136 columns certify). Equalities (2.2) say that the 30 generators f_j kill all homology in homological degree one of $S[e]$, and that the classes of the 136 cycles $d(g_a)$ span $\ker D_2 / \operatorname{im} D_3$, the homology in homological degree two of $S[e,f]$ after the decomposable boundaries. *No assertion that the* 136 *columns span, or are independent in, H^3 of any cochain complex is made anywhere in this paper, and no total dimension of any T^3 or H^3 group is used.* The role of (2.2) is: (i) it validates the generator table (2.1) as a genuine initial Tate segment, so that a full resolution continues with generators of homological degree ≥ 4 only; (ii) it supplies the two cycle-lifting hypotheses used in Lemma 3.3 and Lemma 5.1.

2.2 Which dg Lie algebra controls what

Two strict dg Lie algebras act on P, and the proof must distinguish them. With the conventions of §1.2 put

$$\mathfrak{g} = \operatorname{Der}_k(P,P)_0, \qquad \mathfrak{h} = \operatorname{Der}_S(P,P)_0.$$

The *relative* algebra $\mathfrak{h}$ consists of derivations vanishing on $S = P_0$; perturbing d by a degree-one element of $\mathfrak{h}$ changes nothing about the polynomial ring or its coordinates. It is therefore the controller of *embedded deformations in the fixed coordinates* $x_1, \dots, x_8$, which is what Theorem 1.1 is about. The *absolute* algebra $\mathfrak{g}$ also contains degree-zero coordinate vector fields; its H^1 is the abstract tangent space computed by the packages. The two are compared, never identified:

Lemma 2.4. $\mathfrak{h}^p = \mathfrak{g}^p$ *for every* $p \geq 1$*; consequently* $Z^p(\mathfrak{h}) = Z^p(\mathfrak{g})$ *and* $B^p(\mathfrak{h}) \subseteq B^p(\mathfrak{g})$ *for* $p \geq 1$*, and*

$$H^1(\mathfrak{h}) \twoheadrightarrow H^1(\mathfrak{g}), \qquad H^q(\mathfrak{h}) = H^q(\mathfrak{g}) \quad (q \geq 2). \tag{2.3}$$

Proof. A derivation of cohomological degree $p \geq 1$ maps $P_0 = S$ into homological degree $-p < 0$, which is zero; so it is automatically S-linear, giving $\mathfrak{h}^p = \mathfrak{g}^p$. For $p \geq 1$ the differentials $\partial : \mathfrak{h}^p \to \mathfrak{h}^{p+1}$ and $\partial : \mathfrak{g}^p \to \mathfrak{g}^{p+1}$ coincide, so the cocycles agree in all degrees ≥ 1 and the coboundary spaces agree in all degrees ≥ 2 (both are $\partial(\mathfrak{g}^1)$). In degree one, $B^1(\mathfrak{h}) = \partial(\mathfrak{h}^0) \subseteq \partial(\mathfrak{g}^0) = B^1(\mathfrak{g})$, whence the surjection. *The surjection* $H^1(\mathfrak{h}) \to H^1(\mathfrak{g})$ *is far from injective*: Computation 2.7 gives dimensions 109 and 53. □

2.3 Tangent identifications

Let $C^\bullet = \mathrm{Der}_k(P, A)_0$ and $C^\bullet_S = \mathrm{Der}_S(P, A)_0$, with differential $\partial\theta = -(-1)^p \theta d$ (the differential of A is zero). Since A is concentrated in homological degree zero, a degree-p cochain is determined by its values on the degree-p generators of (2.1); explicitly, in weight zero,

$$C^1_S = C^1 = (A_3)^{16}, \quad C^2_S = C^2 = (A_4)^{30}, \quad C^3_S = C^3 = (A_5)^{16} \oplus (A_6)^{120}, \tag{2.4}$$

while $C^0 = (A_1)^8$ (values on the x_i) and $C^0_S = 0$.

Lemma 2.5 (derivation quasi-isomorphisms)**.** *Composition with the augmentation $P \to A$ induces quasi-isomorphisms*

$$\mathfrak{g} = \mathrm{Der}_k(P, P)_0 \longrightarrow C^\bullet, \qquad \mathfrak{h} = \mathrm{Der}_S(P, P)_0 \longrightarrow C^\bullet_S. \tag{2.5}$$

Proof. We give the absolute case; the relative case is identical with $\Omega_{P/S}$ in place of $\Omega_{P/k}$. The dg P-module $\Omega_{P/k}$ is free as a graded P-module on the basis $\{dx_i\} \cup \{dz : z \text{ a Tate generator}\}$, and $\mathrm{Der}_k(P, M) = \mathrm{Hom}_P(\Omega_{P/k}, M)$ for every dg P-module M. Filter $\Omega_{P/k}$ by the sub-dg-modules F_q generated by the dx_i and the dz with z of homological degree $\leq q$. For a generator z of degree q, $d(dz) = d(d_P z)$ is the universal derivative of an element of the subalgebra on earlier generators, hence lies in F_{q-1}; so each F_q is d-stable and F_q/F_{q-1} is a free P-module on classes with zero induced differential. Let $Q = \ker(P \to A)$; it is acyclic, because $P \to A$ is a surjective quasi-isomorphism. For each q, $\mathrm{Hom}_P(F_q/F_{q-1}, Q)$ is a product of shifted copies of Q, hence acyclic (homology commutes with direct products of complexes of vector spaces), and by induction each $\mathrm{Hom}_P(F_q, Q)$ is acyclic (a degreewise-split extension of acyclic complexes). The restrictions $\mathrm{Hom}_P(F_q, Q) \to \mathrm{Hom}_P(F_{q-1}, Q)$ are surjective because $F_{q-1} \subset F_q$ splits as graded modules; hence in the inverse limit $\mathrm{Hom}_P(\Omega_{P/k}, Q) = \varprojlim_q \mathrm{Hom}_P(F_q, Q)$ the $\varprojlim^1$-terms of the Milnor sequence vanish and the limit is acyclic. Finally $0 \to \mathrm{Hom}(\Omega, Q) \to \mathrm{Hom}(\Omega, P) \to \mathrm{Hom}(\Omega, A) \to 0$ is exact because Ω is graded-free, so the second map is a quasi-isomorphism. All modules split as direct products over internal weights, homology commutes with those products, and therefore the weight-zero components in (2.5) are quasi-isomorphisms as well. □

Lemma 2.6 (cohomology in low degrees)**.** *With identifications* (2.4)*:*

1. $H^1(\mathfrak{h}) = H^1(C_S) = \mathrm{Hom}_S(I, A)_0$*, the space of weight-zero S-module maps $I \to A$, via* $\theta \mapsto (m_i \mapsto \theta(e_i))$*.*

2. $H^1(\mathfrak{g}) = H^1(C) = \mathrm{Hom}_S(I, A)_0/\mathrm{im}(\mathrm{Jac})$*, where* Jac *sends a weight-zero vector field* $\xi = \sum \xi_i \partial/\partial x_i$ *to* $(m_i \mapsto \xi(m_i))$*. This is the classical embedded-to-abstract quotient defining* $T^1_0(A)$*.*

3. $H^2(\mathfrak{h}) = H^2(\mathfrak{g}) = T^2_0(A)$ *and* $H^3(\mathfrak{h}) = H^3(\mathfrak{g})$*, computed in* $C^\bullet$ *as* $\ker \delta^2/\mathrm{im}\, \delta^1$ *and* $\ker \delta^3/\mathrm{im}\, \delta^2$ *respectively, where δ^1 is (the transpose of) R_0 read in A and δ^2 is the linear f-part of the $d(g_a)$ read in A.*

Proof. (1) $C^0_S = 0$, so $H^1(C_S) = \ker(\delta^1)$; a degree-one cochain with values $v_i = \theta(e_i) \in A_3$ is closed iff $\theta(df_j) = \sum_i (R_0)_{ij} v_i = 0$ in A for all j. Because R_0 generates *all* first syzygies of $(m_1, \dots, m_{16})$ (Computation 2.2), this is exactly the condition that $m_i \mapsto v_i$ is a well-defined S-map $I \to A$. (2) The absolute complex has the extra term $C^0 = (A_1)^8$ with $\partial\xi$ acting on e_i by $\mp\xi(m_i)$; the image is the Jacobian image, and the sign does not change it. (3) For $q \geq 2$ the two cochain complexes agree from degree one onward except in degree zero and one, and coboundaries in degree ≥ 2 come from degree ≥ 1, where the complexes agree; combine with Lemma 2.5. A degree-two derivation kills the e_i (their values would land in negative homological degree of A) and hence kills the decomposables $e_i e_j$; so $\delta^2\eta(g_a) = \pm\eta(dg_a)$ sees only the linear f-part of dg_a. The identification of $H^2(C)$ with the package's $T^2_0(A)$, at the level of the specific bases used, is part of Computation 5.3 below. □

Verified computation 2.7 (controller dimensions)**.** The exact script `verify_fable_dgla.m2` in this directory computes

$$\dim \mathrm{Hom}_S(I, A)_0 = 109, \qquad \dim \mathrm{im}(\mathrm{Jac})_0 = 56, \qquad \dim T^1_0(A) = 53 = 109 - 56,$$

and $\dim T^2_0(A) = 27$. The certificate is `verification/fable_dgla_QQ.txt`. Thus $H^1(\mathfrak{h})$ has dimension 109 and $H^1(\mathfrak{g})$ dimension 53; the proof never conflates them.

3 Maurer–Cartan elements give embedded flat graded families

For $n \geq 1$ set $\mathfrak{h}_n = \mathfrak{h} \otimes_k (s)/(s^n)$, a nilpotent dg Lie algebra over R_n, and let

$$\widehat{\mathfrak{h}} = s\,\mathfrak{h}[[s]] = \prod_{j \geq 1} s^j \mathfrak{h}, \qquad F^r \widehat{\mathfrak{h}} = s^r \mathfrak{h}[[s]], \tag{3.1}$$

a dg Lie algebra with a complete, separated filtration for which brackets add filtration levels. The Maurer–Cartan sets are

$$\mathrm{MC}(\mathfrak{h}_n) = \left\{ \alpha \in \mathfrak{h}^1_n : \ \partial\alpha + \tfrac{1}{2}[\alpha, \alpha] = 0 \right\}, \qquad \text{and likewise } \mathrm{MC}(\widehat{\mathfrak{h}}).$$

Proposition 3.1 (realization)**.** *Let $\alpha \in \mathrm{MC}(\mathfrak{h}_n)$. Then $D = d + \alpha$ is an $R_n[x]$-linear, weight-preserving, square-zero derivation of $P \otimes_k R_n$ of homological degree -1, and:*

1. *$H_i(P \otimes R_n, D) = 0$ for $i > 0$;*
2. *$B_n := H_0(P \otimes R_n, D) = R_n[x_1, \dots, x_8]/(D(e_1), \dots, D(e_{16}))$, and each $D(e_i)$ is a homogeneous cubic congruent to m_i modulo s;*
3. *every weight component $(B_n)_j$ is a free R_n-module of rank $\dim_k A_j$; in particular B_n is a flat graded embedded deformation of A over R_n in the fixed coordinates.*

The same statements hold over R for $\alpha \in \mathrm{MC}(\widehat{\mathfrak{h}})$, with (3) replaced by: each weight component of $B = H_0(P \otimes R, D)$ satisfies $B/s^n B = B_n$ for all n with each $(B_n)_j$ free.

Proof. For a degree-one derivation α one has $(d+\alpha)^2 = \frac{1}{2}[d+\alpha, d+\alpha] = \partial\alpha + \frac{1}{2}[\alpha, \alpha]$ (using $d^2 = 0$), so the Maurer–Cartan equation is literally $D^2 = 0$. S-linearity of α and of d gives $R_n[x]$-linearity of D; weight preservation holds because $\alpha \in \mathfrak{h}_0$ by definition.

(1) Induct on n. The sub-complex $s^{n-1}(P \otimes R_n)$ is D-stable and, since $\alpha \cdot s^{n-1} \equiv 0$, is isomorphic to (P, d); the quotient is $(P \otimes R_{n-1}, D \bmod s^{n-1})$. The long exact homology sequence and the inductive hypothesis (base $n = 1$: (P, d) itself) give $H_i = 0$ for $i > 0$ and the exact rows

$$0 \to A \to H_0(P \otimes R_n, D) \to H_0(P \otimes R_{n-1}, D) \to 0. \tag{3.2}$$

(2) $(P \otimes R_n)_0 = R_n[x]$ and $(P \otimes R_n)_1$ is the free $R_n[x]$-module on $e_1, \dots, e_{16}$; by $R_n[x]$-linearity the image of D in degree zero is the ideal generated by the $D(e_i)$. Each $D(e_i) = m_i + \alpha(e_i)$ has weight 3, i.e. is a cubic in x with coefficients in $(s) \subset R_n$ beyond its leading term m_i.

(3) Fix a weight j. Each $(P \otimes R_n)_{i,j}$ is a finite free R_n-module: there are finitely many Tate generators in each homological degree (Lemma 2.1), each of weight ≥ 3, so a fixed pair (i, j) admits only finitely many generator monomials, each with a finite-dimensional space of polynomial coefficients. By (1) and (2),

$$\cdots \to (P \otimes R_n)_{1,j} \xrightarrow{D} (P \otimes R_n)_{0,j} \to (B_n)_j \to 0$$

is a resolution of $(B_n)_j$ by finite free R_n-modules. Applying $- \otimes_{R_n} k$ returns the complex $(P_{*,j}, d)$, because $D \equiv d \pmod{s}$; its positive homology vanishes. Hence $\mathrm{Tor}_1^{R_n}(k, (B_n)_j) = H_1(P_{*,j}, d) = 0$. Now use the local flatness criterion for the finite module $(B_n)_j$ over the Artinian local ring R_n: choose a surjection $G \to (B_n)_j$ from a finite free module with $G \otimes k \cong (B_n)_j \otimes k$ and kernel $Z \subseteq (s)G$; vanishing of Tor_1 makes $0 \to Z \otimes k \to G \otimes k \to (B_n)_j \otimes k \to 0$ exact, so $Z \otimes k = 0$ and $Z = 0$ by Nakayama. (This is the Artinian case of [6, Tag 00MK].) Thus $(B_n)_j$ is free; its rank equals $\dim_k (B_n)_j \otimes_{R_n} k$, and since H_0 is a cokernel and hence commutes with the base change $R_n \to k$, this is $\dim_k A_j$.

Over R: reduction of D modulo s^n is the realization of $\alpha \bmod s^n \in \mathrm{MC}(\mathfrak{h}_n)$, and $H_0(P \otimes R, D)/s^n \cong H_0(P \otimes R_n, D_n)$ in each weight by (3.2) and right-exactness; the displayed statements follow. □

Remark 3.2. No converse ("every deformation comes from a Maurer–Cartan element") and no equivalence of deformation functors is invoked anywhere in this paper. The one deformation we must convert into a Maurer–Cartan element—the certified family over R_4—is converted by the explicit construction of Section 4. This is why no deformation-functor formalism, in dg Lie or any other language, appears among the dependencies.

The extension tool for that construction is:

Lemma 3.3 (filtered cycle lifting)**.** *Let $Q \subseteq P$ be a d-stable graded subalgebra containing S and all Tate generators of homological degree $< q$, where $q \geq 3$, and suppose $H_{q-2}(Q, d) = 0$. Let $D = d + \beta$ be a square-zero weight-preserving differential on $Q \otimes R_n$ with β S-linear of positive s-order. Then every weight-homogeneous d-cycle $c_0 \in Q_{q-1}$ admits a weight-homogeneous D-cycle $c \in Q_{q-1} \otimes R_n$ with $c \equiv c_0 \pmod{s}$.*

Proof. Build $c^{(r)} = c_0 + su_1 + \cdots + s^{r-1}u_{r-1}$ with $Dc^{(r)} \equiv 0 \pmod{s^r}$; the case $r = 1$ is $dc_0 = 0$. Write $Dc^{(r)} \equiv s^r \rho \pmod{s^{r+1}}$ with $\rho \in Q_{q-2}$. Applying D and using $D^2 = 0$ gives $s^r d\rho \equiv 0 \pmod{s^{r+1}}$, so $d\rho = 0$. Since $H_{q-2}(Q) = 0$, choose $u_r \in Q_{q-1}$ with $du_r = -\rho$, homogeneous of the required weight (both d and D preserve weight, so ρ is weight-homogeneous and the certified graded syzygy computations allow a homogeneous primitive). Then $c^{(r+1)}$ improves the congruence; after $n - 1$ steps $c = c^{(n)}$ is a D-cycle. □

4 The certified R_4 family is a genuine truncated Maurer–Cartan element

4.1 The certified matrices

Fix the rational tangent vector, in the ordered 53-column basis T^1 computed by `CT^1(0,F)`:

$$y = (1, 3, 1, 2, 6, 1, 1, 1, 1, 1, 21, 28, 1, 5, 10, 55, 66, 15, 1, 18, 2,\\ 35, 42, 1, 4, 10, 1, 25, 12, 30, 5, 33, 44, 1, 6, 3, 14, 7, 1, 9, 1, 6,\\ 12, 20, 1, 4, 22, 1, 15, 1, 1, 8, 11). \tag{4.1}$$

No property of the provenance of (4.1) is used.

Verified computation 4.1 (the starting family)**.** Write $F_0, \dots, F_3$ and $Q_0, \dots, Q_3$ for the order-0 to order-3 family and relation-lift matrices returned by the pinned `versalDeformation` run (`HighestOrder=>3`, `SmartLift=>false`), and specialize the parameters along $t = sy$. Put $F^{[3]} = \sum_{i\le 3} F_i$ and $Q^{[3]} = \sum_{i\le 3} Q_i$ over $R_4[x_1, \dots, x_8]$. The exact scripts `../completion/verify_starting_jet.m2` and `verify_fable_dgla.m2` verify:

1. the quadratic and cubic base equations vanish at y, and
$$F^{[3]}\, Q^{[3]} = 0 \quad \text{over } R_4[x]; \tag{4.2}$$
2. Q_0 is the *complete* 30-column first-syzygy matrix of the row $(m_1, \dots, m_{16})$;
3. *weight homogeneity*: every entry of $F^{[3]}$ is homogeneous of x-degree 3 and every entry of $Q^{[3]}$ is homogeneous of x-degree 1 (new check in `verify_fable_dgla.m2`);
4. the first-order coefficient of $F^{[3]}$ equals $T^1 y$ entrywise;
5. the reduction of $F^{[3]}$ modulo s^3 regenerates, byte for byte, the frozen sixteen-row two-jet file with SHA-256

`40e64e61674b6a4e61f1ea6822dc79327bf4ba397285f84ed8738c4c18cd1795`.

Certificates: `../completion/verification/starting_jet_QQ.txt`, `verification/fable_dgla_QQ.txt`.

4.2 Lifting the matrices to a square-zero Tate differential

The matrices of Computation 4.1 are *not* treated as cohomology classes. They are used as the values of an explicit derivation.

The Tate model's relation matrix R_0 (Lemma 2.1) and the package syzygy matrix Q_0 generate the same module; the frozen script `../referee-packet/code/verify_aq_bracket.m2` computes (and every replay reverifies) a constant 30×30 matrix U with

$$Q_0 U = R_0, \qquad \det U = 1. \tag{4.3}$$

Proposition 4.2 (the truncated Maurer–Cartan element)**.** *Define a degree -1, weight-preserving, $R_4[x]$-linear derivation D on $P \otimes R_4$ by*

$$D(e_i) = F_i^{[3]}, \qquad D(f_j) = \sum_{i=1}^{16} (Q^{[3]}U)_{ij}\, e_i, \tag{4.4}$$

and on the Tate generators of homological degree ≥ 3 by Lemma 3.3, as detailed in the proof. Then $D^2 = 0$, $D \equiv d \pmod{s}$, *and*

$$\bar{\alpha} := D - d \;\in\; \mathrm{MC}(\mathfrak{h} \otimes (s)/(s^4)). \tag{4.5}$$

Moreover $H_0(P \otimes R_4, D) = R_4[x]/(F^{[3]}) =: A_4$, *a flat graded embedded deformation whose reduction modulo* s^3 *is the printed two-jet, and the linear coefficient* a *of* $\bar{\alpha}$ *is a weight-zero cocycle in* $\mathfrak{h}^1$ *with* $a(e_i) = (T^1 y)_i$.

Proof. On the generators (4.4): $D^2(e_i) = D(F_i^{[3]}) = 0$ by $R_4[x]$-linearity, and

$$D^2(f_j) = \sum_i (Q^{[3]}U)_{ij} F_i^{[3]} = \left(F^{[3]} Q^{[3]} U\right)_j = 0$$

by (4.2). Weight preservation on e, f is Computation 4.1(3). Reduction modulo s: $F^{[3]} \equiv (m_i)$ and $Q^{[3]}U \equiv Q_0 U = R_0$, so $D \equiv d$ on e and f.

Extend over the remaining generators by induction on homological degree. For a generator z of degree $q \geq 3$, the element dz is a weight-homogeneous d-cycle of degree $q-1$ in the subalgebra $P_{<q}$ on the earlier generators. The hypothesis $H_{q-2}(P_{<q}, d) = 0$ of Lemma 3.3 holds: for $q = 3$ it is $\ker D_1 = \operatorname{im} D_2$ and for $q = 4$ it is $\ker D_2 = \operatorname{im}[D_3\, Z]$, both certified in (2.2); for $q \geq 5$ it holds because $P_{<q} = P^{(q-1)}$ is the Tate stage with $H_i = 0$ for $1 \leq i \leq q-2$ (Lemma 2.1). Also D restricted to $P_{<q} \otimes R_4$ squares to zero by the induction (its square, an even derivation, vanishes on generators). Lemma 3.3 produces a weight-homogeneous D-cycle $c_z \equiv dz \pmod{s}$; set $D(z) = c_z$. Then $D^2(z) = D(c_z) = 0$, and $D \equiv d \pmod{s}$ on z. The extension of a value set from generators to all of $P \otimes R_4$ by the signed Leibniz rule is a derivation, and the square of the odd derivation D is the derivation $\frac{1}{2}[D, D]$, so $D^2 = 0$ everywhere.

Consequently $\bar{\alpha} = D - d$ is S-linear (both D and d vanish on $S \otimes R_4$, being $R_4[x]$-linear by definition), weight-zero, and of degree one with coefficients in $(s)/(s^4)$, and $D^2 = 0$ is (4.5). Proposition 3.1 identifies $H_0 = R_4[x]/(D(e_i)) = A_4$ and gives flatness; the reduction of $F^{[3]}$ modulo s^3 is the printed two-jet by Computation 4.1(5). The Maurer–Cartan equation modulo s^2 reads $\partial a = 0$, and $a(e_i)$ is the s-coefficient of $F_i^{[3]}$, which is $(T^1 y)_i$ by Computation 4.1(4). □

5 The exact sequence at H^2

5.1 The computed derivations and their global closure

The frozen script `../referee-packet/code/verify_aq_bracket.m2` constructs, by exact linear algebra over $\mathbf{Q}$: a degree-one derivation θ with $\theta(e_i) = (T^1 y)_i$, solved to be closed on all generators (2.1); degree-one derivations $\psi_1, \ldots, \psi_{53}$ with $\psi_t(e_i) = (T^1)_{it}$, solved to be closed through the f_j; and degree-two derivations $\eta_1, \ldots, \eta_{27}$ whose f-values are the transport by U^T of the package T^2 basis, checked to be closed on all g_a.

Lemma 5.1 (global extension of the computed derivations)**.** *θ and each η_j extend, without changing their values on* (2.1), *to closed derivations of the full Tate resolution P; each ψ_t extends without changing its values on the e_i, f_j.*

Proof. Let ϑ be a derivation of cohomological degree $p \in \{1, 2\}$, closed on all generators of homological degree $< q$. Closure on a generator z of degree q demands $d(\vartheta z) = (-1)^p \vartheta(dz)$; the right side is a d-cycle (apply closure on earlier generators to $d(dz) = 0$) of homological degree $q - p - 1$. For the first open case of ψ_t, namely $p = 1$, $q = 3$, that degree is 1 and $\ker D_1 = \operatorname{im} D_2$ (2.2) supplies a weight-homogeneous primitive in $P_{<3}$. For all cases with $q \geq 4$ the degree $q-p-1$ is positive and lies in the stage $P^{(q-1)}$, which is acyclic there (Lemma 2.1), so a primitive among earlier generators exists; choose its component of the required weight. Induction over the well-ordered generators finishes. The values already checked are never altered. $\square$

Corollary 5.2. *The* 27 *commutators* $[\theta, \eta_j] = \theta\eta_j - \eta_j\theta$ *are closed degree-three derivations of* P*; their images in* C^3 *lie in* $\ker\delta^3$*. Hence, on their span, the projection*

$$H^3(\mathfrak{g}) = \ker\delta^3 / \operatorname{im}\delta^2 \hookrightarrow C^3 / \operatorname{im}\delta^2 \tag{5.1}$$

computes rank in genuine $H^3(\mathfrak{g})$*. Likewise the* 53 *commutators* $[\theta, \psi_t] = \theta\psi_t + \psi_t\theta$ *are closed degree-two derivations, representing classes of* $H^2(\mathfrak{g})$*.*

This is the rigorous content behind the "136 cycles" step: the 136 generators enter only through the certified completeness (2.2), which (i) makes the displayed g-stage the correct domain for degree-three cochain values, and (ii) guarantees the closure solutions and extensions above. Nothing about a basis of H^3 is claimed.

Verified computation 5.3 (bracket data)**.** The frozen script, replayed exactly, proves over $\mathbf{Q}$:

1. the transported 27 columns are ∂-closed, independent modulo $\operatorname{im}\delta^1$, and no weight-zero cocycle in C^2 remains outside their span plus $\operatorname{im}\delta^1$; hence $[\eta_1], \ldots, [\eta_{27}]$ is a basis of the 27-dimensional $H^2(\mathfrak{g})$;
2. the matrix of $\big([\theta, \eta_j]\big)_{j\leq 27}$ in $C^3/\operatorname{im}\delta^2$ has rank 12;
3. the matrix of $\big([\theta, \psi_t]\big)_{t\leq 53}$ in $C^2/\operatorname{im}\delta^1$ equals $-J\,Dq_y$, where J is the (injective modulo boundaries) matrix of the 27 classes and Dq_y is the Jacobian at y of the 27 quadratic package equations; rank $Dq_y = 15$;
4. the composite of the matrices in (2) and (3) is zero, and the column $([\theta, \psi_t])y$ vanishes.

Certificate: `../referee-packet/verification/aq_bracket_QQ.txt` (replayed lines: `rank commutator columns modulo delta2=12`, `rank Dq_y=15`, `primary plus Dq rank=0`, `ad_y composed primary-bracket rank=0`, `Jacobi composite rank=0`).

5.2 From the script's cocycle to the actual linear coefficient

Let a be the linear coefficient of the constructed element (4.5) and θ the script's cocycle.

Lemma 5.4 (chain-level bridge)**.** $a - \theta = \partial\xi$ *for some* $\xi \in \mathfrak{g}^0$*. Consequently, for every closed* $z \in \mathfrak{h}^p$ *with* $p \geq 1$*,*

$$[a, z] - [\theta, z] = \partial[\xi, z], \qquad [\xi, z] \in \mathfrak{h}^p,$$

so ad_a *and* ad_θ *induce the same maps* $H^p(\mathfrak{h}) \to H^{p+1}(\mathfrak{h})$ *for all* $p \geq 1$*.*

Proof. Both a and θ are closed ($\partial a = 0$ from the Maurer–Cartan equation modulo s^2; θ by construction), and both take the value $(T^1 y)_i \in S_3$ on e_i, *literally as polynomials* (Proposition 4.2 and the definition of θ). A degree-one element of $C^1 = \operatorname{Der}_k(P, A)_0$ is determined by its e-values; hence the image of $a - \theta$ in C^1 is the zero cochain. By Lemma 2.5 the map $H^1(\mathfrak{g}) \to H^1(C)$ is injective, so $[a - \theta] = 0$ in $H^1(\mathfrak{g})$, i.e. $a - \theta = \partial\xi$ with $\xi \in \mathfrak{g}^0$. For closed z of degree $p \geq 1$, the derivation property of ∂ gives $\partial[\xi, z] = [\partial\xi, z] + (-1)^0[\xi, \partial z] = [a - \theta, z]$, and $[\xi, z]$ has cohomological degree $p \geq 1$, hence lies in $\mathfrak{h}^p = \mathfrak{g}^p$ (Lemma 2.4). $\square$

5.3 Exactness

Proposition 5.5 (exactness at H^2)**.** *With a the linear coefficient of* (4.5)*,*

$$H^1(\mathfrak{h}) \xrightarrow{\mathrm{ad}_a} H^2(\mathfrak{h}) \xrightarrow{\mathrm{ad}_a} H^3(\mathfrak{h}) \tag{5.2}$$

is exact at the 27*-dimensional middle term; the first map has image of dimension* 15 *and the second has rank* 12*.*

Proof. By Lemma 5.4 it suffices to prove the statement for ad_θ. We first prove it for the absolute algebra $\mathfrak{g}$ and then transfer.

Absolute exactness. By Computation 5.3(1) and Lemma 2.6(3), the $[\eta_j]$ are a basis of $H^2(\mathfrak{g})$. By Corollary 5.2 and (5.1), the rank 12 of Computation 5.3(2) is the rank of $\mathrm{ad}_\theta : H^2(\mathfrak{g}) \to H^3(\mathfrak{g})$; note that this uses the injection (5.1) only on already-closed columns, so no dimension of $H^3(\mathfrak{g})$ is needed or claimed. By Computation 5.3(3), the image of $\mathrm{ad}_\theta : H^1(\mathfrak{g}) \to H^2(\mathfrak{g})$ is the column space of $J\,Dq_y$; since the 27 columns of J are independent modulo boundaries, this image has dimension $\operatorname{rank} Dq_y = 15$. By Computation 5.3(4) the composite vanishes on the computed matrices, i.e. $\operatorname{im}\mathrm{ad}_\theta \subseteq \ker \mathrm{ad}_\theta$ at H^2. Finally

$$\dim\ker\bigl(\mathrm{ad}_\theta : H^2 \to H^3\bigr) = 27 - 12 = 15 = \dim\operatorname{im}\bigl(\mathrm{ad}_\theta : H^1 \to H^2\bigr),$$

so the inclusion is an equality. *The exactness is thus obtained from the two independently computed ranks together with the directly verified zero composite—not from the single rank of Dq_y.*

Transfer to $\mathfrak{h}$. By Lemma 2.4, $H^2(\mathfrak{h}) = H^2(\mathfrak{g})$ and $H^3(\mathfrak{h}) = H^3(\mathfrak{g})$, and the second map in (5.2) is the absolute one; so $\ker(\mathrm{ad}_\theta : H^2(\mathfrak{h}) \to H^3(\mathfrak{h}))$ has dimension 15. For the image of the first map: the surjection $H^1(\mathfrak{h}) \to H^1(\mathfrak{g})$ of Lemma 2.4 intertwines the two maps ad_θ, so $\mathrm{ad}_\theta\, H^1(\mathfrak{h}) \supseteq$ the image computed from any absolute class represented by a relative cocycle—and every absolute class is so represented since $Z^1(\mathfrak{h}) = Z^1(\mathfrak{g})$. Conversely if a relative class dies absolutely, i.e. is represented by $\partial\xi$ with $\xi \in \mathfrak{g}^0$, then

$$[\theta, \partial\xi] = -\partial[\theta, \xi],$$

because $\partial\theta = 0$ and ∂ is a degree-one derivation of the bracket, and $[\theta, \xi] \in \mathfrak{g}^1 = \mathfrak{h}^1$; so such classes contribute nothing to the image in $H^2(\mathfrak{h})$. Hence $\operatorname{im}(\mathrm{ad}_\theta : H^1(\mathfrak{h}) \to H^2(\mathfrak{h})) = \operatorname{im}(\mathrm{ad}_\theta : H^1(\mathfrak{g}) \to H^2(\mathfrak{g}))$, of dimension 15, and (5.2) is exact at H^2. □

Remark 5.6*.* The 56 extra relative tangent classes of Computation 2.7 are precisely the coordinate directions that become absolute boundaries; the proof above shows they are invisible to the obstruction calculus. This is why the 109/53 discrepancy—which would be fatal if the two H^1's were confused—is harmless when tracked correctly.

6 The strict all-orders induction

The abstract lemma is stated and proved in full, with all signs, in the companion note `DG_LIE_INDUCTION.tex`; we reproduce it here so that this document is standalone.

For $\gamma \in \widehat{\mathfrak{h}}^1$ define the curvature

$$\mathcal{F}(\gamma) = \partial\gamma + \tfrac{1}{2}[\gamma, \gamma]. \tag{6.1}$$

Lemma 6.1 (strict Bianchi identity)**.** $\partial\mathcal{F}(\gamma) + [\gamma, \mathcal{F}(\gamma)] = 0.$

Proof. Since ∂ is a degree-one derivation of the bracket and $|\gamma| = 1$: $\partial[\gamma, \gamma] = [\partial\gamma, \gamma] - [\gamma, \partial\gamma] = -2[\gamma, \partial\gamma]$, using $[\partial\gamma, \gamma] = -(-1)^{2\cdot 1}[\gamma, \partial\gamma]$. Hence $\partial\mathcal{F}(\gamma) = -[\gamma, \partial\gamma]$. On the other hand $[\gamma, \mathcal{F}(\gamma)] = [\gamma, \partial\gamma] + \frac{1}{2}[\gamma, [\gamma, \gamma]]$, and the graded Jacobi identity applied to three copies of the odd element γ gives $3[\gamma, [\gamma, \gamma]] = 0$, hence $[\gamma, [\gamma, \gamma]] = 0$ over $\mathbf{Q}$. Adding the two displays gives the identity. $\square$

Lemma 6.2 (fixed-two-jet extension). *Let $\bar{\alpha} \in \mathrm{MC}(\mathfrak{h} \otimes (s)/(s^4))$ have linear coefficient a, and suppose* (5.2) *is exact at $H^2(\mathfrak{h})$. Then there is*

$$\alpha_\infty \in \mathrm{MC}(\widehat{\mathfrak{h}}) \qquad \textit{with} \qquad \alpha_\infty \equiv \bar{\alpha} \pmod{s^3}. \tag{6.2}$$

Proof. Choose any lift $\gamma^{(4)} \in \widehat{\mathfrak{h}}^1$ of $\bar{\alpha}$ (for instance with no coefficients beyond s^3); then $\mathcal{F}(\gamma^{(4)}) \in F^4\widehat{\mathfrak{h}}$ because the Maurer–Cartan equation holds modulo s^4. Inductively assume, for $N \geq 4$, an element $\gamma^{(N)} = sa + O(s^2)$ with

$$\mathcal{F}(\gamma^{(N)}) = s^N r_N + s^{N+1} r_{N+1} + O(s^{N+2}), \qquad r_i \in \mathfrak{h}^2. \tag{6.3}$$

Step 1: the leading obstruction is a cocycle. The coefficient of s^N in the Bianchi identity of Lemma 6.1 is $\partial r_N = 0$ (all bracket terms pair a curvature coefficient, of order $\geq N$, with a coefficient of $\gamma^{(N)}$, of order ≥ 1).

Step 2: its class lies in $\ker \mathrm{ad}_a$. The coefficient of s^{N+1} in the Bianchi identity is

$$\partial r_{N+1} + [a, r_N] = 0,$$

so $[a, r_N]$ is exact and $[r_N] \in \ker(\mathrm{ad}_a : H^2(\mathfrak{h}) \to H^3(\mathfrak{h}))$.

Step 3: kill it by a closed degree-one correction plus a boundary. By exactness (Proposition 5.5) there is a *closed* $v \in Z^1(\mathfrak{h})$ with $[r_N] = -[[a, v]]$ in $H^2(\mathfrak{h})$; choose $w \in \mathfrak{h}^1$ with

$$r_N + [a, v] = \partial w. \tag{6.4}$$

Set $\gamma^{(N+1)} = \gamma^{(N)} + \delta$ with $\delta = s^{N-1}v - s^N w$. The exact change formula (valid in any dg Lie algebra, for $|\delta| = 1$) is

$$\mathcal{F}(\gamma + \delta) - \mathcal{F}(\gamma) = \partial\delta + [\gamma, \delta] + \tfrac{1}{2}[\delta, \delta]. \tag{6.5}$$

Here $\partial\delta = -s^N \partial w$ (as $\partial v = 0$); $[\gamma^{(N)}, \delta] = s^N[a, v] + O(s^{N+1})$, because $\gamma^{(N)} - sa = O(s^2)$ and $[\gamma^{(N)}, s^N w] = O(s^{N+1})$; and $\frac{1}{2}[\delta, \delta] = O(s^{2N-2})$ with $2N - 2 \geq N + 2$ for $N \geq 4$. So the coefficient of s^N changes by $[a, v] - \partial w = -r_N$ by (6.4), no lower coefficient changes, and $\mathcal{F}(\gamma^{(N+1)}) = O(s^{N+1})$, restoring (6.3) with $N + 1$.

Step 4: convergence and the fixed two-jet. The step indexed by N changes γ only in s-orders $\geq N - 1$; hence for each fixed j the coefficient of s^j stabilizes after step $N = j + 1$, and by completeness of the filtration (3.1) the sequence converges coefficientwise to some $\alpha_\infty \in \widehat{\mathfrak{h}}^1$. For each j, the s^j-coefficient of $\mathcal{F}(\alpha_\infty)$ is a finite expression in the first j coefficients of α_∞, which agree with those of $\gamma^{(N)}$ for $N > j + 1$; since $\mathcal{F}(\gamma^{(N)}) \in F^N\widehat{\mathfrak{h}}$, that coefficient is zero. Separatedness ($\bigcap_N F^N = 0$) gives $\mathcal{F}(\alpha_\infty) = 0$. The first correction, at $N = 4$, is $s^3 v - s^4 w$; therefore no coefficient of s or s^2 is ever changed and (6.2) holds. $\square$

Remark 6.3 (the initial case, explicitly). With the lift $\gamma^{(4)} = sa_1 + s^2 a_2 + s^3 a_3$ (the coefficients of $\bar{\alpha}$, $a_1 = a$), the curvature is exactly

$$\mathcal{F}(\gamma^{(4)}) = s^4\Big([a_1, a_3] + \tfrac{1}{2}[a_2, a_2]\Big) + s^5[a_2, a_3] + s^6\tfrac{1}{2}[a_3, a_3],$$

all lower coefficients vanishing because $\bar{\alpha}$ is Maurer–Cartan over $(s)/(s^4)$. So $r_4 = [a_1, a_3] + \frac{1}{2}[a_2, a_2]$, and Step 1 for $N = 4$ asserts $\partial r_4 = 0$, which also follows by direct expansion.

Remark 6.4 (what is and is not preserved). The conclusion fixes $\bar{\alpha}$ modulo s^3, *not* modulo s^4: the first correction $s^3 v$ may alter a_3. Retaining the whole R_4-element would need the strictly stronger hypothesis $[r_4] = 0$ in $H^2(\mathfrak{h})$, which exactness does not provide and which we do not claim. Everything downstream (Section 8) uses only the coefficients of s^0, s^1, s^2, i.e. only the printed two-jet, so the weaker preservation suffices.

Combining Proposition 4.2, Proposition 5.5, and Lemma 6.2:

Corollary 6.5. *There is* $\alpha_\infty \in \mathrm{MC}(s\,\mathfrak{h}[[s]])$ *with* $\alpha_\infty \equiv \bar{\alpha} \pmod{s^3}$.

7 The algebraic family over $\mathbf{Q}[[s]]$

Put $D_\infty = d + \alpha_\infty$ on $P \otimes R$ and define

$$F_i := D_\infty(e_i) \in R[x_1, \ldots, x_8] \qquad (1 \le i \le 16). \tag{7.1}$$

Lemma 7.1. *Each F_i is an honest polynomial: homogeneous of degree three in $x_1, \ldots, x_8$ with coefficients in $R = \mathbf{Q}[[s]]$, and*

$$F_i = F_i^{(2)} + s^3 H_i, \qquad H_i \in R[x]_3, \tag{7.2}$$

where $F_1^{(2)}, \ldots, F_{16}^{(2)}$ are exactly the printed frozen two-jet cubics.

Proof. α_∞ preserves weight, so $\alpha_\infty(e_i)$ has weight three: each s-coefficient lies in the finite-dimensional space S_3, making F_i an element of $R \otimes S_3 = R[x]_3$ (no completion in x occurs). By Corollary 6.5 and Computation 4.1(5), $F_i \equiv F_i^{[3]} \equiv F_i^{(2)} \pmod{s^3}$. □

Define

$$B = R[x_1, \ldots, x_8]/(F_1, \ldots, F_{16}), \qquad X = \operatorname{Proj} B \subset \mathbf{P}^7_R. \tag{7.3}$$

Proposition 7.2 (flat projective family)**.** *$X \to \operatorname{Spec} R$ is a projective, flat morphism with special fibre $X \times_R k = X_0$. Moreover every weight component B_j is a free R-module of rank $\dim_k A_j$; in particular the Hilbert functions of all fibres coincide with that of A.*

Proof. X is a closed subscheme of $\mathbf{P}^7_R$ cut by homogeneous polynomials, so it is projective over R; no algebraization theorem is required, because (7.1) are already polynomials (Lemma 7.1).[1] The special fibre is $\operatorname{Proj}(S/(F_i \bmod s)) = \operatorname{Proj}(S/I) = X_0$.

For each n, the reduction $\alpha_\infty \bmod s^n$ is a Maurer–Cartan element of $\mathfrak{h}_n$ whose realization is $D_\infty \bmod s^n$; by Proposition 3.1,

$$B/s^n B = R_n[x]/(F_i \bmod s^n) = H_0(P \otimes R_n, D_\infty \bmod s^n)$$

has all weight components free over R_n of rank $\dim_k A_j$. Fix j; B_j is a finitely generated R-module (a quotient of the finite free module $R[x]_j$). If $sb = 0$ for some $b \in B_j$, then for every n the image of b in the free R_n-module $B_j/s^n B_j$ is killed by s, hence lies in $s^{n-1}(B_j/s^n B_j)$, so $b \in s^{n-1} B_j$; the Krull intersection theorem for the finite module B_j over the Noetherian local ring R forces $b = 0$. A finitely generated torsion-free module over the discrete valuation ring R is free; its rank is $\dim_k B_j/sB_j = \dim_k A_j$. Hence $B = \bigoplus_j B_j$ is R-flat. On the standard chart $\{x_i \neq 0\}$ the coordinate ring $(B_{x_i})_0$ is a direct summand of the localization B_{x_i} of the flat module B, hence flat; the charts cover X, so $X \to \operatorname{Spec} R$ is flat. □

[1] If one prefers to phrase the family formally first, the compatible flat closed subschemes $X_n = \operatorname{Proj}(B/s^n B) \subset \mathbf{P}^7_{R_n}$ satisfy every hypothesis of [7, Tag 0899]: R is Noetherian and (s)-adically complete, $\mathbf{P}^7_R \to \operatorname{Spec} R$ is separated and of finite type, the squares are cartesian closed immersions, and $X_1 = X_0$ is proper over $\mathbf{Q}$. The resulting algebraization is the closed subscheme (7.3), by the uniqueness part applied to the explicit cubics. We do not need this route.

8 The generic fibre: Cohen–Macaulay, equidimensional, smooth

8.1 Cohen–Macaulayness and equidimensionality

Let $B_K = B \otimes_R K = K[x]/(F_1, \dots, F_{16})K[x]$ be the homogeneous coordinate ring of the generic fibre and, for a field extension L/K, put $B_L = B_K \otimes_K L$.

Lemma 8.1 (Betti semicontinuity across the family)**.** *For all i,* $\dim_K \mathrm{Tor}_i^{K[x]}(B_K, K) \le \dim_k \mathrm{Tor}_i^S(A, k)$.

Proof. Choose a graded free resolution $L_\bullet \to B$ by finitely generated free graded $R[x]$-modules ($R[x]$ is Noetherian and B is a finitely generated graded module). Every syzygy module $Z_i = \ker(L_i \to L_{i-1})$ is R-flat: inductively, in $0 \to Z_i \to L_i \to Z_{i-1} \to 0$ both right terms are R-flat ($Z_{-1} = B$ by Proposition 7.2), so $\mathrm{Tor}_1^R(Z_{i-1}, -) = 0$ keeps the sequence exact after any base change and Z_i is flat as a kernel of a surjection of flats onto a flat. Consequently $L_\bullet \otimes_R k$ is exact except at the end, i.e. it is a graded free resolution of A over $S = k[x]$, and likewise $L_\bullet \otimes_R K$ resolves B_K over $K[x]$.

Let $N_\bullet = L_\bullet \otimes_{R[x]} R[x]/(x_1, \dots, x_8)$, a complex of finitely generated free R-modules. Then $H_i(N_\bullet \otimes_R k) = \mathrm{Tor}_i^S(A, k)$ and $H_i(N_\bullet \otimes_R K) = \mathrm{Tor}_i^{K[x]}(B_K, K)$ by the previous paragraph. Over the discrete valuation ring R, for a complex of finite free modules, $\dim_K H_i(N \otimes K) = \mathrm{rank}_R H_i(N)$ while universal coefficients give an injection $H_i(N) \otimes k \hookrightarrow H_i(N \otimes k)$; hence $\dim_K H_i(N \otimes K) \le \dim_k H_i(N) \otimes k \le \dim_k H_i(N \otimes k)$. ☐

Proposition 8.2. *For every field extension L of K, the ring B_L is Cohen–Macaulay of Krull dimension 4, and all its minimal primes p satisfy* $\dim B_L/p = 4$*. Consequently the geometric generic fibre $X_{\bar K} = \mathrm{Proj}\, B_{\bar K}$ is Cohen–Macaulay with every irreducible component of dimension exactly 3, and every local ring of $X_{\bar K}$ at a closed point has Krull dimension 3.*

Proof. The Hilbert function: $\dim_L (B_L)_j = \dim_K (B_K)_j = \mathrm{rank}_R B_j = \dim_k A_j$ (Proposition 7.2). Since A has Krull dimension 4, its Hilbert polynomial has degree 3, and by Hilbert–Serre dimension theory [2, Ch. 4] the same holds for B_L: $\dim B_L = 4$.

Projective dimension: $\mathrm{depth}_{S_+} A = 4$ (Computation 1.2), so the graded Auslander–Buchsbaum formula [2, §1.3] gives $\mathrm{pd}_S A = 8 - 4 = 4$, i.e. $\mathrm{Tor}_i^S(A, k) = 0$ for $i \ge 5$. By Lemma 8.1, $\mathrm{Tor}_i^{K[x]}(B_K, K) = 0$ for $i \ge 5$, so $\mathrm{pd}_{K[x]} B_K \le 4$; since Tor commutes with the flat base change $K \to L$, also $\mathrm{pd}_{L[x]} B_L \le 4$. Auslander–Buchsbaum again gives $\mathrm{depth}_{\mathfrak m} B_L \ge 8 - 4 = 4$ where $\mathfrak m$ is the irrelevant maximal ideal; combined with $\dim B_L = 4$ and depth $\le$ dim, the local ring $(B_L)_{\mathfrak m}$ is Cohen–Macaulay of dimension 4, and hence so is the graded ring B_L (Cohen–Macaulayness of a finitely generated graded algebra over a field is detected at the irrelevant maximal ideal, and localization preserves it) [2, §2.1].

Unmixedness: a Cohen–Macaulay local ring is equidimensional—every minimal prime q of $(B_L)_{\mathfrak m}$ has $\dim (B_L)_{\mathfrak m}/q = \dim (B_L)_{\mathfrak m}$ [2, §2.1]. The minimal primes of B_L are homogeneous, hence contained in $\mathfrak m$ and in bijection with those of $(B_L)_{\mathfrak m}$, and graded dimensions are computed at $\mathfrak m$; so $\dim B_L/p = 4$ for every minimal prime. No minimal prime contains $(B_L)_+$ (its quotient would have dimension $\le 1 < 4$), so the irreducible components of $\mathrm{Proj}\, B_L$ are the $\mathrm{Proj}(B_L/p)$, of dimension $4 - 1 = 3$. At a closed point of $X_{\bar K}$, the local ring has dimension equal to the maximum dimension of components through the point, namely 3; since all components have dimension 3, this holds at every closed point. ☐

8.2 Singularity as a linear-algebra incidence

Work on the affine chart $x_v = 1$ of $\mathbf{P}^7$, $1 \le v \le 8$, with coordinates $z_1, \dots, z_7$, and let J denote the 16×7 Jacobian matrix $(\partial \widetilde{F}_i / \partial z_u)$ of the dehomogenized cubics. Let $\bar K$ be an algebraic closure of K.

Lemma 8.3 (Jacobian dichotomy)**.** *A closed point p of $X_{\bar K}$ lying in the chart $x_v = 1$ is a singular point of $X_{\bar K}$ if and only if* $\operatorname{rank} J(p) \le 3$*, if and only if* $\ker J(p) \subseteq \bar K^7$ *contains a four-dimensional subspace.*

Proof. The chart $X_{\bar K} \cap \{x_v \neq 0\}$ is the affine scheme cut out by exactly the sixteen dehomogenized cubics: for a graded quotient $B_{\bar K} = \bar K[x]/(F_i)$, the degree-zero part of the localized ideal $\big((F_i)_{x_v}\big)_0$ is generated by the F_i/x_v^3, since any g/x_v^m with $g = \sum h_i F_i$ homogeneous of degree m equals $\sum (h_i/x_v^{m-3})(F_i/x_v^3)$. Hence the Zariski tangent space of $X_{\bar K}$ at p is $\ker J(p)$, of dimension $7 - \operatorname{rank} J(p) \ge 3$. By Proposition 8.2, $\dim \mathcal{O}_{X_{\bar K},p} = 3$. The point is smooth iff $\mathcal{O}_{X_{\bar K},p}$ is regular iff $\dim_{\bar K} \ker J(p) = 3$ iff $\operatorname{rank} J(p) = 4$; it is singular iff $\operatorname{rank} J(p) \le 3$ iff $\dim \ker J(p) \ge 4$. □

For a four-subset $\Pi \subset \{1, \dots, 7\}$ let K_Π be the 7×4 matrix whose rows indexed by Π form the identity and whose remaining twelve entries are indeterminates $a_1, \dots, a_{12}$; these are the 35 standard charts of $\mathrm{Gr}(4,7)$. The *truncated incidence generators* on the pair (v, Π) are the 80 polynomials

$$\widetilde F_1^{(2)}, \dots, \widetilde F_{16}^{(2)}, \quad (J^{(2)})\, K_\Pi \;\in\; \mathbf{Q}[z_1, \dots, z_7, a_1, \dots, a_{12}][s], \tag{8.1}$$

where the superscript (2) denotes the printed two-jet cubics and their Jacobian ($J^{(2)} K_\Pi$ has $16 \cdot 4$ entries; the frozen scripts label it via the transpose convention, which changes nothing).

Verified computation 8.4 (the 280 incidence memberships)**.** For every one of the $8 \cdot 35 = 280$ pairs (v, Π), the frozen scripts `../referee-packet/code/verify_incidence_chart.m2` and `../referee-packet/code/verify_all_incidence.py` prove by exact rational module Gröbner reduction the membership

$$s^2 \in (q_1, \dots, q_{80}) + (s^3) \subset \mathbf{Q}[z, a][s], \tag{8.2}$$

where $q_1, \dots, q_{80}$ are the generators (8.1): the 80 generators are encoded over $\mathbf{Q}[z,a][s]/(s^3)$ as a 3×240 module matrix and $(0,0,1)^{\mathsf T}$ is exactly reduced to zero against it. 277 pairs certify at degree bound six and pairs $(2,7), (6,12), (8,16)$ at degree bound eight; the driver checks that all 280 pairs occur and re-verifies one retained lift. Replayed line: `proved 280/280 chart pairs`. Certificate: `../referee-packet/verification/incidence_all_QQ.txt`.

What this does and does not certify: (8.2) is a family of 280 polynomial identities about the *two-jet* (8.1) only. It makes no reference to the higher-order coefficients H_i, and it is not, by itself, a smoothness statement; the geometric content is extracted in Proposition 8.5.

8.3 Smoothness of the geometric generic fibre

Proposition 8.5. *The geometric generic fibre $X_{\bar K}$ is smooth. Indeed, this holds for every flat projective family whose equations satisfy* (7.2) *with the printed two-jet.*

Proof. Suppose $X_{\bar K}$ has a singular closed point. By Lemma 8.3, on some chart $x_v = 1$ there is a pair (p, Λ) of a point and a four-plane $\Lambda \subseteq \ker J(p)$. The incidence scheme

$$\Sigma = \Big\{(p, \Lambda) \in (X \cap \{x_v \neq 0\}) \times \mathrm{Gr}(4,7) :\; J(p) \cdot \Lambda = 0\Big\}$$

(over all v) is a scheme of finite type over K, and it has a $\bar K$-point; hence it is nonempty and has a closed point whose residue field K' is a finite extension of K (Hilbert Nullstellensatz for finite-type schemes over a field).

Let R' be the integral closure of $R = \mathbf{Q}[[s]]$ in K'. Since R is a complete discrete valuation ring, R' is again a complete discrete valuation ring, finite over R [5, Ch. II]; write v' for its valuation, so $v'(s) > 0$.

The point p is a K'-point of the projective scheme X; by the valuative criterion of properness it extends to an R'-point of $X \subset \mathbf{P}^7_{R'}$. Scale its homogeneous coordinates to lie in R' with at least one of minimal valuation a unit; call that coordinate x_{v_0}. The chart may change ($v_0 \neq v$ is possible), so re-choose the plane there: p lies in both charts, the Zariski tangent space $T_pX_{K'}$ is intrinsic, and each chart is cut out by its sixteen dehomogenized cubics (proof of Lemma 8.3), so $\dim_{K'} \ker J_{v_0}(p) = \dim_{K'} T_pX_{K'} = \dim_{K'} \ker J_v(p) \geq 4$, the last inequality because $J_v(p)\Lambda = 0$ with $\dim \Lambda = 4$. Choose a four-plane $\Lambda' \subseteq \ker J_{v_0}(p)$ over K'. As a K'-point of the projective scheme $\mathrm{Gr}(4,7)$, Λ' likewise extends to an R'-point; scaling its Plücker coordinates to R' with a unit entry places it in a standard chart Π_0, where it is represented by a matrix K_{Π_0} with the twelve entries in R', pivot rows the identity, and column span reducing to Λ' over K'. The incidence entries $J_{v_0} K_{\Pi_0}$, zero over K' because the columns lie in $\ker J_{v_0}(p)$, lie in R' and hence vanish in R', since $R' \hookrightarrow K'$.

Fix the pair (v_0, Π_0) and let $(\zeta, \alpha) \in (R')^7 \times (R')^{12}$ be these integral coordinates. Membership (8.2), lifted to the polynomial ring, is an identity

$$s^2 = \sum_{j=1}^{80} c_j\, q_j + s^3 c_0 \qquad \text{in } \mathbf{Q}[z,a][s], \tag{8.3}$$

with explicit polynomials c_j, c_0. By (7.2), the actual incidence generators of the family—the dehomogenized F_i and the entries of $J_{v_0} K_{\Pi_0}$—differ from the truncated q_j by s^3 times polynomials in z, a with coefficients in R: dehomogenizing and differentiating $s^3 H_i$ with respect to a z_u preserves the factor s^3. Every actual generator vanishes at (ζ, α): the dehomogenized F_i vanish because the R'-section lies in X and the chart ideal is generated by them (proof of Lemma 8.3), and the entries of $J_{v_0} K_{\Pi_0}$ vanish by the previous paragraph. Substituting $q_j = q'_j - s^3 \rho_j$ into (8.3) and evaluating at (ζ, α) therefore yields

$$s^2 = s^3 c, \qquad c \in R'.$$

Taking valuations: $2v'(s) = 3v'(s) + v'(c) \geq 3v'(s)$, contradicting $v'(s) > 0$. Hence no singular closed point exists, and $X_{\bar{K}}$, being of finite type over the algebraically closed field $\bar{K}$ with all closed points smooth, is smooth. □

Proof of Theorem 1.1. Proposition 4.2 converts the certified R_4 family into $\bar{\alpha} \in \mathrm{MC}(\mathfrak{h} \otimes (s)/(s^4))$ with the printed two-jet; Proposition 5.5 establishes exactness of (5.2) at $H^2(\mathfrak{h})$ for its linear coefficient; Lemma 6.2 extends $\bar{\alpha}$ to $\alpha_\infty \in \mathrm{MC}(s\,\mathfrak{h}[[s]])$ without changing the coefficients of s and s^2; Lemma 7.1 and Proposition 7.2 produce the flat projective family (7.3) with special fibre X_0 and equations (7.2); and Proposition 8.5 proves its geometric generic fibre smooth. □

9 Dependency ledger

Step	Status	Where
Special fibre CM, depth = 4	Verified computation + Reisner	Comp. 1.2
Tate stages complete	Verified computation (M2 + Singular)	Comp. 2.2
Controller dimensions 109/56/53, 27	Verified computation	Comp. 2.7
MC $\Rightarrow$ flat embedded family	Directly proved	Prop. 3.1
R_4 family $\Rightarrow$ MC element	Verified computation + directly proved	Comp. 4.1, Prop. 4.2
Exactness at H^2	Verified computation (ranks 12, 15, zero composite, full H^2 basis) + directly proved transfer	Comp. 5.3, Prop. 5.5
All-orders MC element	Directly proved (strict Bianchi induction)	Lem. 6.2
Flat projective family over R	Directly proved	Prop. 7.2
Generic fibre CM equidimensional	Directly proved (Betti semicontinuity + Auslander–Buchsbaum)	Prop. 8.2
Generic smoothness	Verified computation (280 memberships) + directly proved valuation argument	Comp. 8.4, Prop. 8.5

External classical inputs: Reisner's criterion [4]; Tate's construction (re-proved as Lemma 2.1); local flatness criterion [6, Tag 00MK] (re-proved in the case used); Auslander–Buchsbaum, Hilbert–Serre, and unmixedness of Cohen–Macaulay local rings [2]; valuative criterion of properness; integral closure of a complete DVR in a finite extension [5]. The optional footnote route uses [7, Tag 0899]. No statement about structures transferred to cohomology, or about operations of arity ≥ 3, is used, cited, or silently reproduced.

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